Unit 2
Problem Solving and Creativity

UNIT 2:
Topics to be Covered

Thinking

- Components of Thinking
- Nature of Thinking
- Thinking and Reasoning

Problem Solving process

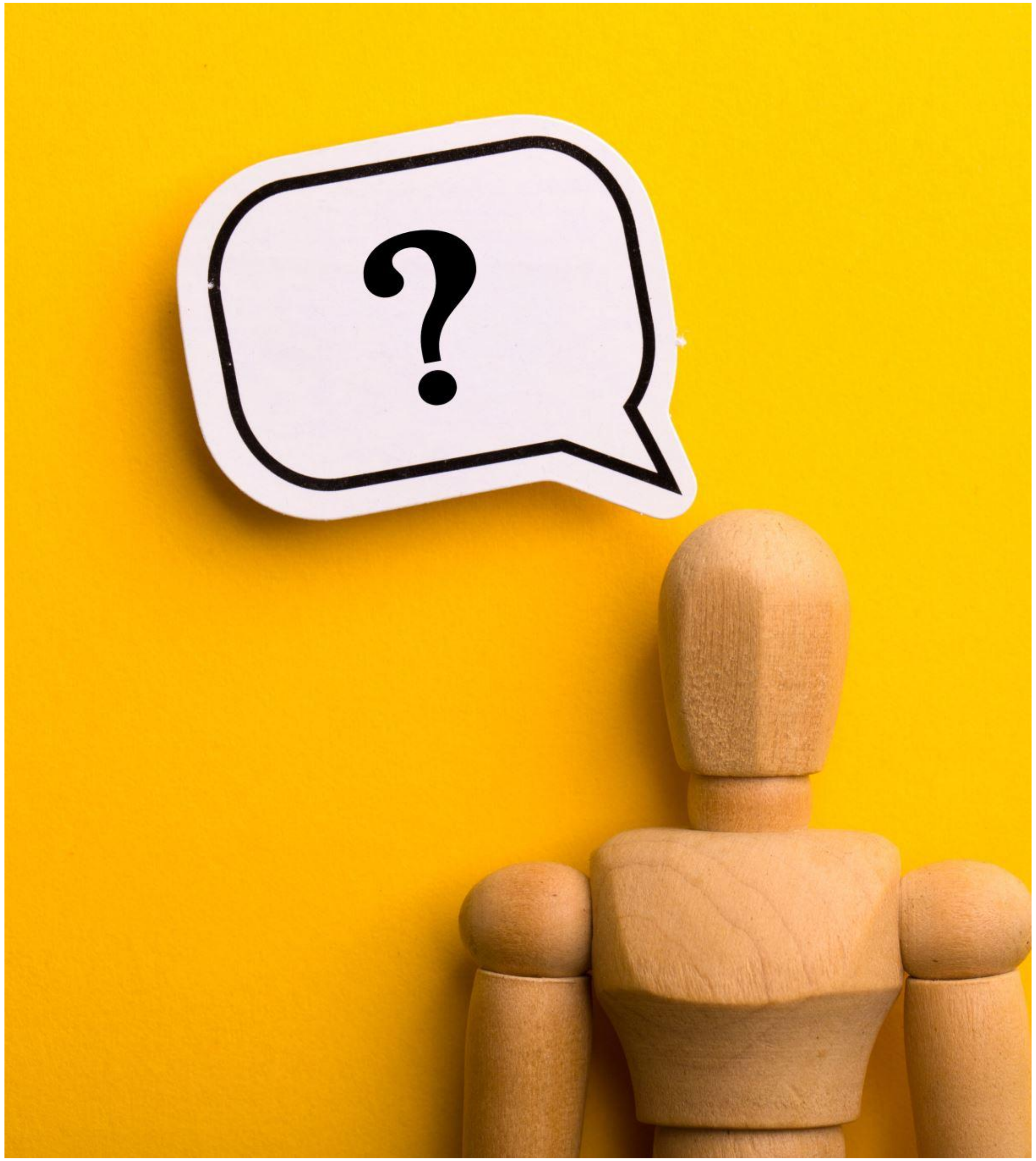
Problem Solving - skills and **Strategies**

Creative Thinking

Model of creative Thinking

Context Setting – thinking

- How many thoughts do you think you’ve already had today? 100? 1,000? Maybe more?”
- Research says the average person has around 60,000 thoughts a day. Wild, right? But here’s the thing—how many of those are you actually aware of?”



Thinking

- What if humans could only communicate through images instead of words?

120 Secs

Possible Responses

•Develop Creative Ideas:

- •New universal set of symbols or
- •Art and design skills for communication
- •Misunderstandings might increase because images can be interpreted in different ways.

•Practical Considerations:

- Difficult to express abstract concepts or emotions using only images.
- Education systems have to change to teach students how to 'read' and 'write' in images.
- Technology, like emojis or GIFs, might evolve for communication

•Philosophical Reflections:

•This could change how we think, making our thought processes more visual rather than verbal.

•It might bring people closer together globally if images transcend language barriers.

•Skeptical or Critical Responses:

•It could limit our ability to communicate complex ideas or emotions."

•"It might be frustrating or confusing if people have different interpretations of the same image."

Mindful Minute (3 to 5 Mins)

Duration per cycle:

- **Inhale** for 4 seconds
- **Hold** for 2 seconds
- **Exhale** for 6 seconds
- **Hold** for 2 seconds
- This makes each cycle last **14 seconds**.

Quick Activity: Word Association - **Fast thinking: Say the First Word Comes to your Mind**

- Thinking
- Innovation
- SRM Campus
- Climate Change
- Canteen
- Problem
- Stress
- Focus
- Decision
- Memory
- Conscious

- Resilience
- Nanotechnology
- Cognitive
- Network
- Mundane
- Illusion
- Serenity
- Curiosity
- Perspective

Quick Activity

- **Take a moment and write down the *last 5 things* you thought about—anything, no filter.**
- Now, look at your list.
- Were they about the past? The future? Yourself? Someone else?”
- That’s thinking in action—and we're going to explore how it works, and how you can take control of it.”

Activity: Fast vs. Slow Thinking (Inspired by Daniel Kahneman)

a) 2 + 2 = ? & 2 * 3 = ?
b) 44 + 120 = ? & 67 * 87 = ?

Discussion:

- The First was *Fast thinking*—automatic, effortless.
- The Second needed *slow thinking*—focused, effortful.

Faculty Note: *When do you think we rely too much on fast thinking, and when is slow thinking actually necessary?*

Fast Thinking

- **Automatic and unconscious:** It happens quickly, without much effort or deliberate thought.
- **Efficient:** This system helps us navigate the world quickly by using shortcuts based on patterns, experiences, and intuition.
- **Reacts to emotions:** It’s influenced by our emotions, instincts, and prior knowledge.
- **Error-prone:** Because it's based on shortcuts, it can lead to errors and mistakes.
- **Examples:**
  - Recognizing a friend's face in a crowd.
  - Reading words on a street sign.
  - Instinctively knowing the answer to 2 + 2.
  - Making a snap judgment about someone

Slow Thinking

- **Deliberate and conscious:** This type of thinking is slower, more effortful, and requires focus.
- **Analytical:** It involves reasoning, logic, and critical thinking to solve problems or make decisions.
- **Consumes energy:** It requires more mental resources, so it’s used less often than fast thinking.
- **More accurate:** Because it's intentional and analytical, it’s less likely to lead to errors (though it can still be flawed).
- **Examples:**
  - Solving a complex math problem (e.g., 17 × 24).
  - Planning a vacation.
  - Evaluating evidence in a debate.
  - Deciding which university to attend based on research

Why It is Important:

- **Fast thinking** is great for everyday tasks and quick decisions but need to be cautious to avoid mistakes or errors.

- **Slow thinking** is essential when we need to solve complex problems, make thoughtful decisions, or engage in critical analysis.

What is Thinking?

- Do we Think while sleeping?
- The difference between what is thinking and what is not thinking is just our awareness about the particular thinking process
- Thinking is usually initiated by a problem and goes through a sequence of steps such as judging, abstracting, inferring, reasoning, imagining, and remembering.
- These steps are often directed towards solution of the problem.

What is thinking? – Definition

- Thinking is the mental process of manipulating information to form **concepts, solve problems, reason, and make decisions**. It involves the use of **cognitive functions** such as **perception, memory, and judgement to process and analyze information**

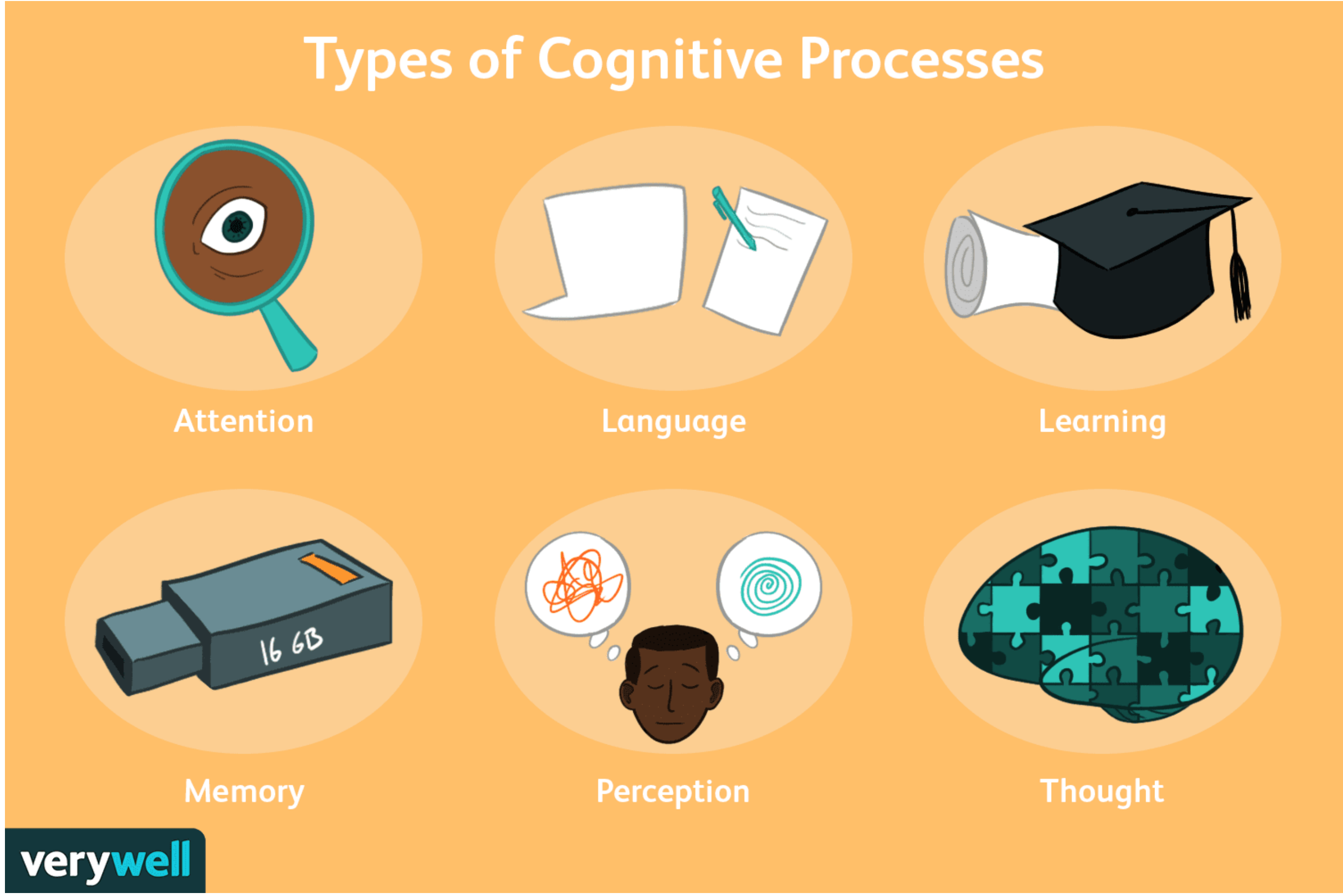
Information Gets collected From

- Collected through senses – eyes, hear, tongue, nose, skin) and from the environment, as well as the information which is stored in our memory because of our encounter with many events and situations in the past.
- Thinking is a constructive process in the sense that it helps us to form a new representation of any object or event by transforming available information.

Thinking and Cognitive activities

- **Thinking is** the base of all cognitive activities or processes and is unique to human beings.

- **Cognitive** –relating to, or being conscious mental activities (as thinking, reasoning, remembering, imagining, learning words and using language)



Concepts and reasoning

- Thinking relies on a variety of mental structures such as concepts and reasoning. We will briefly learn about these mental structures.
- **Concepts:** Concepts are one of the key elements of thinking. Concepts represent objects, activities, ideas, or living organisms.
- They also represent properties (such as “sour” or “brave”), abstractions (such as “anger” or “fear”), and relations (such as “smaller than” or “more intelligent than”). Concepts are mental structures which allow us to organize knowledge in systematic ways. We cannot observe them directly, but we can infer them from behaviour.

Role of Concepts in Thinking

- A concept is a mental representation of a category and refers to a class of objects, ideas or events that share common properties.
- It plays an important role in the thinking process as concept formation helps in organising knowledge so that it can be accessed with less time and effort

Reasoning

- Reasoning is also one of the key aspects of thinking.
- It is a process that involves inference.
- Reasoning is used in logical thinking and problem solving.
- It is goal directed, and the conclusions or judgments are drawn from a set of facts.
- There are two types of reasoning: deductive and inductive.
- **Deductive**: Deduce or draw conclusion from a set of initial assertions or premises (Starting from a general principle and reaching a special conclusion. **Eg: All humans are mortal, Socrates is a human, Therefore Socrates is moral.**
- **Inductive:** we start from available evidence to generate a conclusion about the likelihood of something. Making generalizations based on specific observations. Ex: The sun has risen in the east every day so far. Therefore, the sun will rise in the east tomorrow.

How does reasoning help in solving problems?

- Reasoning helps in problem solving as it is the process of gathering and analysing information to arrive at conclusions. Thus, reasoning helps to arrive at conclusions through certain information. This can be achieved through the following ways:
- i) **Deductive reasoning**: It begins with an assumption that is believed to be true and the conclusion is based on that assumption. Thus, it is reasoning from general to particular.
- (ii) **Inductive reasoning**: It is based on specific facts and observations. It involves the drawing of a general conclusion based on a particular observation.

Take a Minute to answer

- **How did you reach college on time? What were the factors or decisions you have made to reach on time?**

- Find out possible ways of travelling: By Walk / by road / By Train ...
- Factors: Time / shortest route / Easy / convenient / road condition / safety / weather / Not to attend / bike repair/ etc..
- Decision Making is important to act.
- **Requires Thinking for this Simple problems/ activity.**

Fool or Wise do the same things with ONE Difference

- Fool finally do it
- Wise Do it at the beginning

Types of Thinking

Critical	Analytical	Inductive
Deductive	Strategic	Lateral
	Creative	

Nature of Thinking

- Thinking is the basis of all cognitive activities and processes, which are unique to humans. It involves manipulation and analysis of information received from the environment.
- It is the higher mental process through which things are manipulated, and the required information is analysed.
- For example, while playing a video game the mind thinks of strategies or techniques to win it.
- Thus, thinking is organised and goal-directed. It is an internal mental process that is inferred through overt behaviour.

Types of Thinking

- Critical Thinking
  Analyzing facts and evidence carefully before making a judgment or decision.
- Creative Thinking
  Using imagination to come up with new and original ideas or solutions.
- Analytical Thinking
  Breaking down complex information into smaller parts to understand it better.
- Logical Thinking
  Using step-by-step reasoning to solve a problem or make a decision
- Reflective Thinking
  Thinking back on your experiences to learn from them.
- Concrete Thinking
  Focusing on physical objects or specific facts, rather than abstract ideas.
- Abstract Thinking
  Understanding concepts and ideas that are not tied to physical things.
- Lateral Thinking
  Looking at problems from new and different angles to find creative solutions.
- Divergent Thinking
  Generating many different ideas or possibilities from a single starting point.
- Convergent Thinking
  Focusing on finding the single best solution to a clearly defined problem.

What are the various barriers to creative thinking?

- The various barriers to creative thinking are:
  - **Habitual** - The tendency to be overpowered by habits according to a particular think acts as a barrier to creative thinking. It hinders the generation of thought from a fresh perspective.
  - **Perceptual** - It prevents the formation of novel and original ideas.
  - **Motivational** - Lack of motivation acts as a barrier for any thought and action.
  - **Emotional** - Emotions like fear of failure, rejection and negativism lead to negative assumptions and result in incapability to think differently.
  - **Cultural** - It refers to excessive adherence to traditions, expectations, conformity pressures and stereotypes. Cultural block arises due to the fear of being different, tendency to maintain status quo, social pressure, etc

How can creative thinking be enhanced?

- Becoming more aware and sensitive in order to notice and respond to the feelings, sights, sounds and textures around.
- Generating maximum amount of ideas or solutions to a given task, in order to increase the flow of thoughts and choosing the best out of them.
- Using Osborn's 'brainstorming' technique to increase the flexibility of ideas. It involves the idea of thinking freely, without any limitations or pre-conceptions.
- Experience and practice leads to an independent thinking while making judgments.
- Engaging in activities that require the use of imagination and original thinking.

How can creative thinking be enhanced

- Getting a feedback on the proposed solutions and also thinking of solutions others may offer.
- Giving the ideas a chance to incubate.
- Drawing diagrams for easily understanding the solutions and visualising causes or consequences of all the solutions.
- Resisting the temptation of getting immediate rewards.
- Being self-confident, positive and aware of all the defences concerned with the problem.

Critical Thinking: **Puzzle – Solving Relay**

- **6 Member Group**
- **Strategy Time:** 2 minutes to discuss on Your Strategies, Responsibility, Team Efforts, Planning

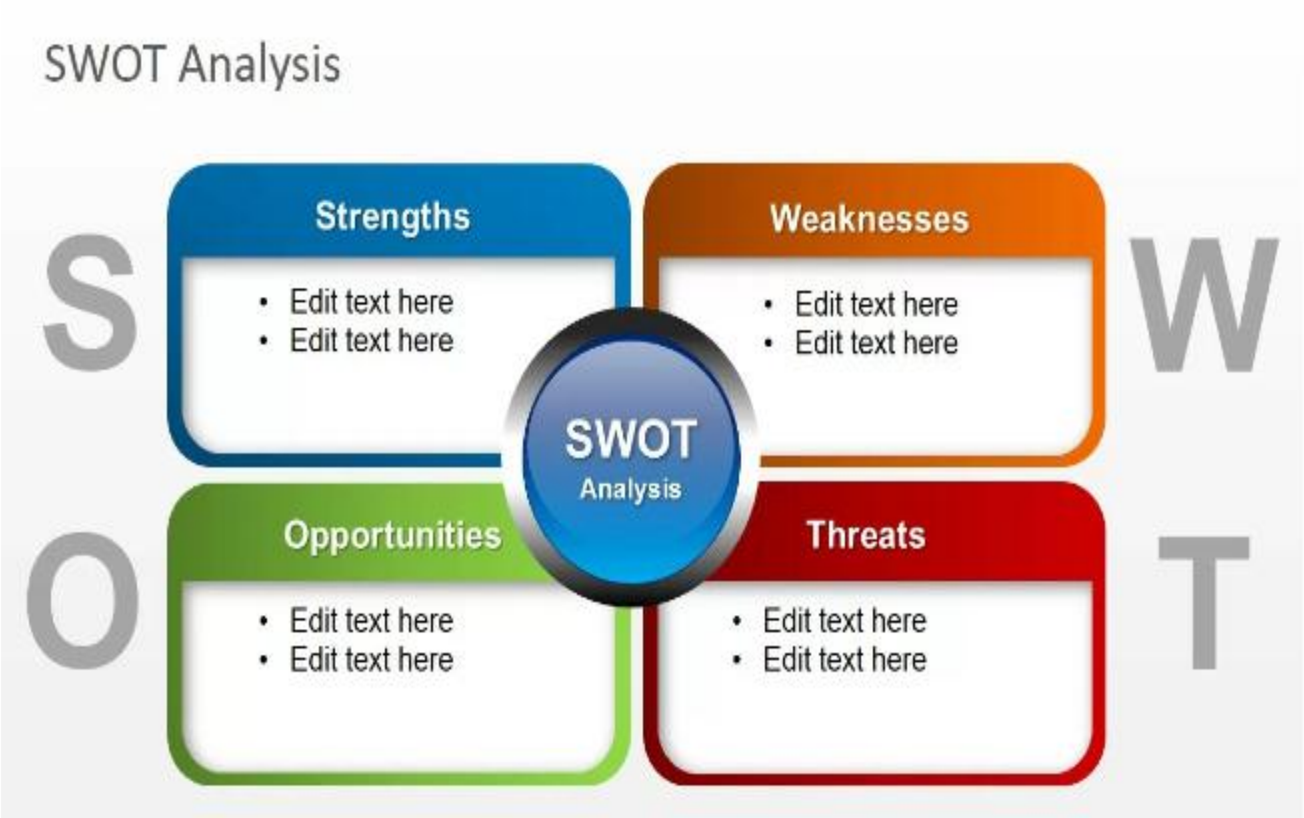
Instruction: 1 puzzle for each member.
Decide which side to start. *Number 1 to 6.*
1 Person – Outline with Right Hand
2 Person – Outline Both hands
3 Person – Outline Left hands
4 Person – Outline Right Hand
5 Person – Outline Both hands
6 Person – Outline Left hands

Task

First Finishing group will be appreciated and rewarded

SWOT – Strength, Weakness, Opportunities and Treats

1. Apply SWOT analysis to assess strengths and weaknesses, Opportunity and Threats
2. Strengths and Weakness are internal
3. Opportunity and Threats are External.



Activity

SWOT Analysis on Individual (5 marks)

1. Time management
2. Health and Wellbeing
3. Build Positive Attitude and behaviour
4. Be Responsible
5. Social Relationship
6. Communication skills
7. Mindful in reducing Screentime

SWOT Analysis on Thematic Topic

- Role of Youth being a Responsible Consumption in reducing greenhouse gas.
- Role of youth in building responsible & ethical robots.
- Role of youth in mitigating drug free college campus.
- Role of youth in reducing single use plastics.

Reverse Thinking: A Fresh Perspective on Problem-Solving & Innovation

- **Reverse thinking**, also known as inversion or working backward, is a problem-solving technique that involves starting with the desired outcome and then identifying the steps or factors that would lead to that outcome.

Key Concepts in Reverse Thinking

- Inverting the Problem : Instead of directly focusing on solutions, reverse thinking encourages identifying the steps that would lead to the opposite of the desired outcome
- **Identifying Limiting Assumptions:** Challenging deeply ingrained beliefs and preconceived notions can unlock new perspectives and ideas.
- **Worst-Case Scenarios:** Analyzing potential failures and negative outcomes can provide valuable insights into the root causes of problems and how to avoid them.
- **Creative Problem Solving:** By re-framing problems and exploring unconventional solutions, reverse thinking can lead to innovative breakthroughs.
- **Adaptability and Resilience:** In a constantly changing world, reverse thinking can help individuals and organizations adapt to uncertainty and unforeseen challenges.

Reverse Thinking Activity: Students

- **1. Time Management Trouble**
- **Situation**: You struggle to finish your work on time and often feel rushed.
  Reverse Thinking Prompt:
  🌀 *If you wanted to completely ruin your time management, what would you do every day?*

- **2. Always Talking, Never Listening**
- **Situation**: You're always talking in class and rarely pay attention when others speak.
  Reverse Thinking Prompt:
  🌀 *If you wanted to be the worst listener in class, what would you do?*

Examples for Reverse thinking

1. Apple

- **Reverse Thinking Approach**: Instead of making devices with keyboards (like Blackberry), Apple asked:
  "What if a phone had no buttons at all?“
- This led to the **iPhone**, changing mobile phones forever.

2. IKEA

- **Reverse Thinking Approach**:
  "What if customers assembled the furniture themselves?"
  Instead of delivering bulky, ready-made furniture, IKEA revolutionized the model with **flat-pack, self-assembly furniture**, lowering costs and enabling global scaling.

3. Netflix

Reverse Thinking Approach: *What if people didn’t have to go to a store to rent movies?"*

Netflix began by mailing DVDs, then asked: *What if there were no physical DVDs at all?"*

That led to the **streaming revolution**, disrupting Blockbuster and traditional media

Wrap ups

- Reverse thinking helps uncover hidden assumptions and encourages innovative solutions by challenging conventional approaches. It fosters creativity, problem-solving, and deeper insights into complex challenges.

Unit – 2
Problem Solving & Skills

In your mind what is a problem?

- A Situation,
- Any Question / Matter Involving Doubt, Mystery, Puzzle, Riddle,
- Unsolved Creates Doubt / Difficulty, Stress,
- A Problem Is Also A Question Raised To Inspire Thought
- In Mathematics, A Problem Is A Statement Or Equation That Requires A Solution.

Who all have the ability to solve a problems

- *Humans [Literate / Illiterate]*
- *Animals & Birds*
- *Insects*
- *Everyone has the ability to solve problems.*

Problem Solving

- Everybody States that “A Manager “Or “A Professor” Has To Be A Master At Problem Solving.
- But This Is Not True!
- Every Human Being Faces Multiple Problems In Life That He Must Try To Solve.

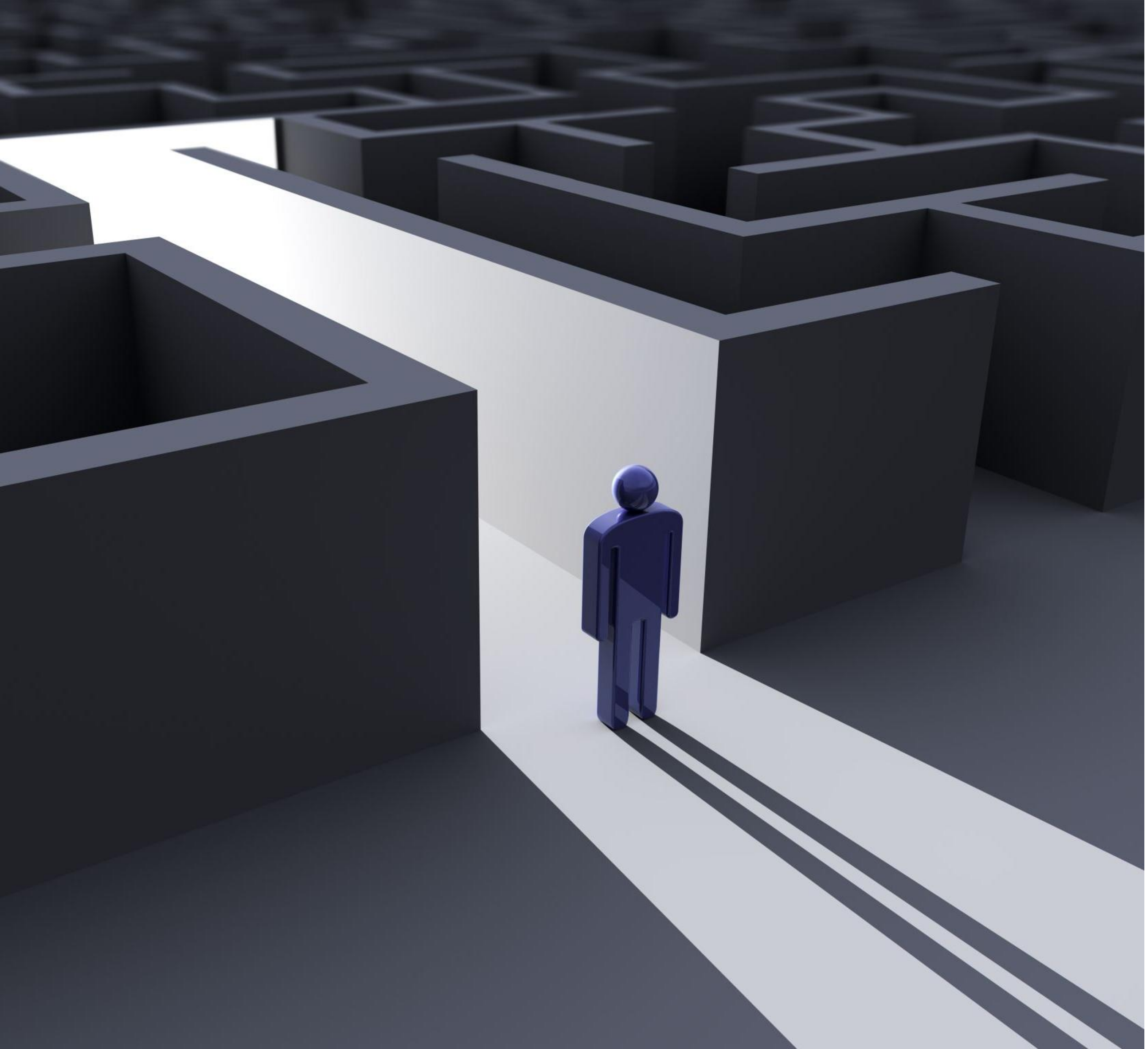
Everyone Has Problem and Solving Ability

Example:

- A housewife has her ability to manage / plan her expenditure
  A Housewife has to plan the expenditure for her house carefully.
- A Student has his/her ability to manage study routine.

What is a Problem Solving?

- **The Process Of Finding A Solution To A Difficult Or Complex Issue.**
- **Involves Logical Reasoning, Creativity, And Critical Thinking.**



Importance of Problem Solving:

- Essential Skill In Academic And Real-world Scenarios.
- Helps In Decision-making, Innovation, And Overcoming Challenges.

Understanding the Nature of Problems

- **What Is A Problem?**
- A Problem Is Any Situation Where **There Is A Gap Between The Current State And A Desired State**, With Obstacles In The Way.
- **Types Of Problems:**
- **Well-defined:** Clear Goals And Solution Paths (Example: Mathematical Problems).
- **Ill-defined:** Unclear Goals And Multiple Possible Solutions (Example: Real-world Challenges).

Why Problem Solving Matters:

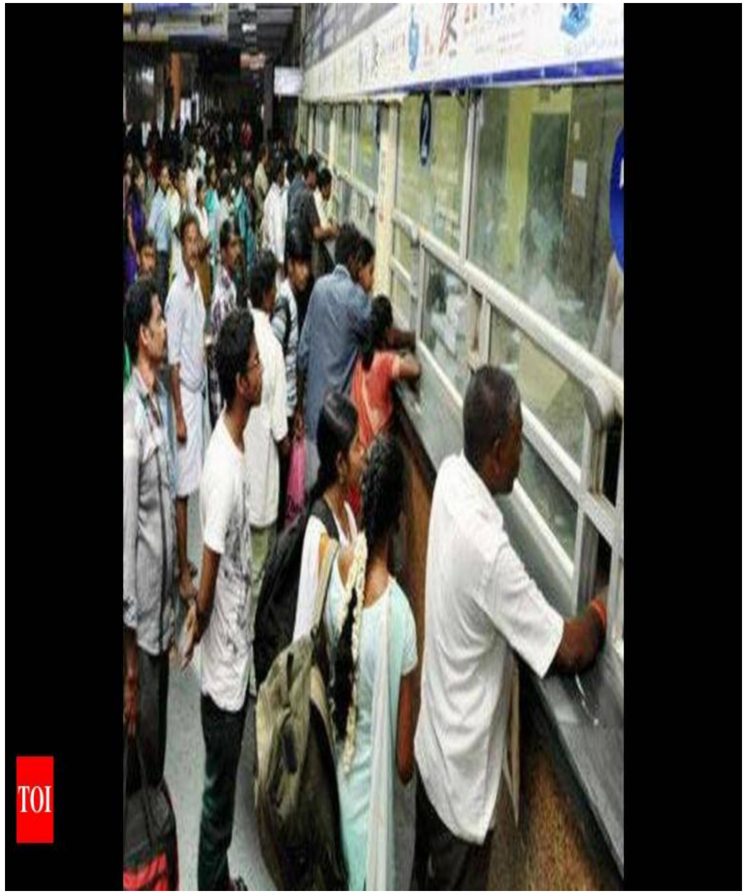
- Crucial Skill For Academic Success And Career Development.
- Enhances Decision-making And Innovation.

Core Problem-Solving Skills Analytical Skills:

- Ability To Break Down Complex Problems Into Manageable Parts.

- **Creative Thinking:**
  - Generating **Innovative And Diverse Solutions.**
- **Decision-making:**
  - Choosing The Best Solution Based On **Evidence And Reasoning.**
- **Resilience:**
  - **Persistence In Overcoming Obstacles** And Learning From Failures.

Do You Guess What Is This?



- Huge Queue For Journey Ticket booking At Railway Station.

  This Problem has been now eased out with Mobile Application & Online Bookings

Problem Solving Process

- **Identify/ understand The Problem**
- **Clarify The Issue:** Understand What The Problem Is And Why It Needs To Be Solved.
- **Gather Information:** Collect Relevant Data And Insights To Understand The Problem's Context.
- **Define The Problem:** Articulate The Problem Clearly, Ensuring Everyone Involved Has A Common Understanding.

Understanding the Problem

1. Clearly **Define The Problem** Before Seeking Solutions.
2. Gather All **Relevant Information And Data**.
3. Identify **The Root Cause**, Not Just The Symptoms.
4. Ask Critical Questions To Uncover Hidden Issues.
5. Develop A Complete Understanding To Guide Effective Solutions.

Define the problem

- Defining The Problem Is The Hardest Part.
- We Usually Tend To Worry About The Problem Before Trying To Identify What The Problem Actually Is.
- We Perceive A Problem, Which Might Not Actually Be The Problem And Start Worrying About It. Instead, Try To Define The Problem

Defining the Problem – 1 – 6 Points

1. What Are The Causes To The Problem?
    - Here You Might Not Identify The Main Causes As Such But It Is A Start. You Have To Keep In Mind That You are not Blaming Somebody As A Cause To The Problem Right At The Beginning.
    - The Person Might Be The Cause, But See What Led To That Person Being The Cause Of The Problem
2. Where Is The Problem Actually Occurring?
3. How Is The Problem Occurring?
4. At What Times Specifically Is The Problem Occurring?
5. Who Is The Problem Happening With?
6. Why Is The Problem Arising? Here You Will Have To Jot Down The Exact Details As To Why The Problem Is Occurring?

The Steps

- Understanding The Problem
- **Critical Thinking:** Think Outside The Box And Consider Multiple Solutions To A Problem
- **Analytical Tools**: SWOT Analysis (Strengths, Weaknesses, Opportunities, Threats
- **Decision Making:** Making Decisions Based On The Information And Analysis Gathered.
- **Implementing Solutions** : Planning, Resource Allocation, And Monitoring The Implementation Process To Ensure Success
- **Reflecting And Learning:** Emphasizes The Importance Of Reflecting On The Outcome Of The Solution And Learning From The Experience
- **Collaboration And Communication** : Stresses The Importance Of Working With Others To Solve Problems

Common fields where problem solving is practiced

- **Psychology:** Problem Solving Is Used In Psychology To Try And Obtain Solutions To Problems Dealing With Mental Health
- **Computer Science And Algorithms:** Every Software Company That Develops New Software Has To Troubleshoot And Solve Problems That The New Software Might Have. In The Field Of Computer Science And Artificial Intelligence Where Algorithms Are The Methods Through Which The Programs Are Designed, Problem Solving Is A Hero!
- **Engineering:** Problem Solving In Engineering Is Used To Overcome Product Or Process Failures. It Is Usually Done To Rectify The Problem And Also To Ensure That The Problem Does Not Occur Again
- **Medicine Science And Health: Research Studies / Public Health.**

Barriers to Problem Solving

1. Problems Are Barriers Themselves
2. Irrelevant Information,
3. Bias Towards Confirmation,
4. Baseless Constraints,
5. Mindset And Fixedness To One Method Of Solving Problems.

Problem Solving Tools & Techniques

1. Root Cause Analysis: Toyota 5 Whys Techniques: Keep asking "Why?" until the root cause is identified.
2. Root cause Analysis: Fishbone Diagram (Ishikawa): Visualizes the cause-and-effect relationship of problems: The **Fishbone Diagram**, also known as the **Ishikawa Diagram** or **Cause-and-Effect Diagram**, is a powerful visual tool used to **identify, explore, and display the possible causes of a specific problem**.
3. SWOT Analysis: **SWOT Analysis** is a structured planning tool used to evaluate the **Strengths, Weaknesses, Opportunities, and Threats** involved in a project, organization, individual, or situation. It helps in **strategic decision-making** by providing a comprehensive view of internal and external factor

Problem Solving Tools & Techniques

4. Brainstorming: **Brainstorming** is a **creative group problem-solving technique** used to generate a large number of ideas or solutions quickly. It emphasizes **quantity over quality initially**, encouraging **free thinking and deferring judgment** to foster innovation and uncover unique perspectives.

5. Flowcharting: **Flowcharting** is a visual method for representing the **steps in a process**, **workflow**, or **system** using standardized symbols. It helps break down complex procedures, analyze inefficiencies, and improve communication and understanding among stakeholders.

Problem Solving Tools & Techniques

6. Pareto Analysis: is a decision-making technique used to identify the most significant factors in a dataset. It’s based on the **Pareto Principle**, also known as the **80/20 rule**, which suggests that **80% of consequences come from 20% of causes**

7. Stakeholder Analysis: **Stakeholder Analysis** is a technique used to **identify, understand, and prioritize all individuals or groups** that are affected by, have influence over, or are interested in a project, policy, or intervention. It’s essential for ensuring **successful planning, communication, and implementation**.

8. Gap Analysis: **GAP Analysis** is a strategic tool used to **compare current performance or state with desired goals or standards** and identify the “gap” that needs to be addressed to achieve those goals

9. Failure Mode and Effects Analysis (FMEA) **Failure Mode and Effects Analysis (FMEA)** is a **systematic, proactive tool** used to identify where and how a process, product, or system might fail and assess the impact of those failures. It helps prioritize risks to improve reliability and safety

Problem Solving Tools & Techniques

10. Six Sigma Tools (DMAIC): A data-driven methodology to improve processes by reducing defects and variation. Uses DMAIC (Define, Measure, Analyze, Improve, Control) framework. Employs various tools at each phase to identify problems, analyze causes, and implement solution.

11. Survey and Feedback Analysis: Collecting information through structured questionnaires or feedback forms.

Analyzing responses to understand attitudes, satisfaction, needs, or performance. Helps in decision-making, program evaluation, and improvement.

What is RCA – Root Cause Analysis

It is a systematic process of discovering the origin of problems or the root cause of the problem in order to determine appropriate solutions. It is the fundamental aspect of problem solving in any field.

- Identify the underlying cause or causes of a problem rather than addressing the immediate symptoms. It ensure that same problem does not occur.

RCA – Root Cause Analysis

Step1: Problem must be clearly and understood [By gathering all data, documenting the data, understanding its impact on the process or system)

Step 2: Asking series of Why Questions – A method known as 5 Whys. The goal of asking this 5 Whys is to peel the symptoms and find out the root cause.

Step 3: One the root cause is been identified solutions can be addressed

Step 4: Develop solutions

Step 5: Implement and Monitor the Solutions

- Root Cause Analysis is not only just fixing problem but about
- Learning Opportunity
- Document Valuable Insights.
- How systems works and how it can be improved.
- Culture of Learning and Improvement

RCA – Root Cause Analysis

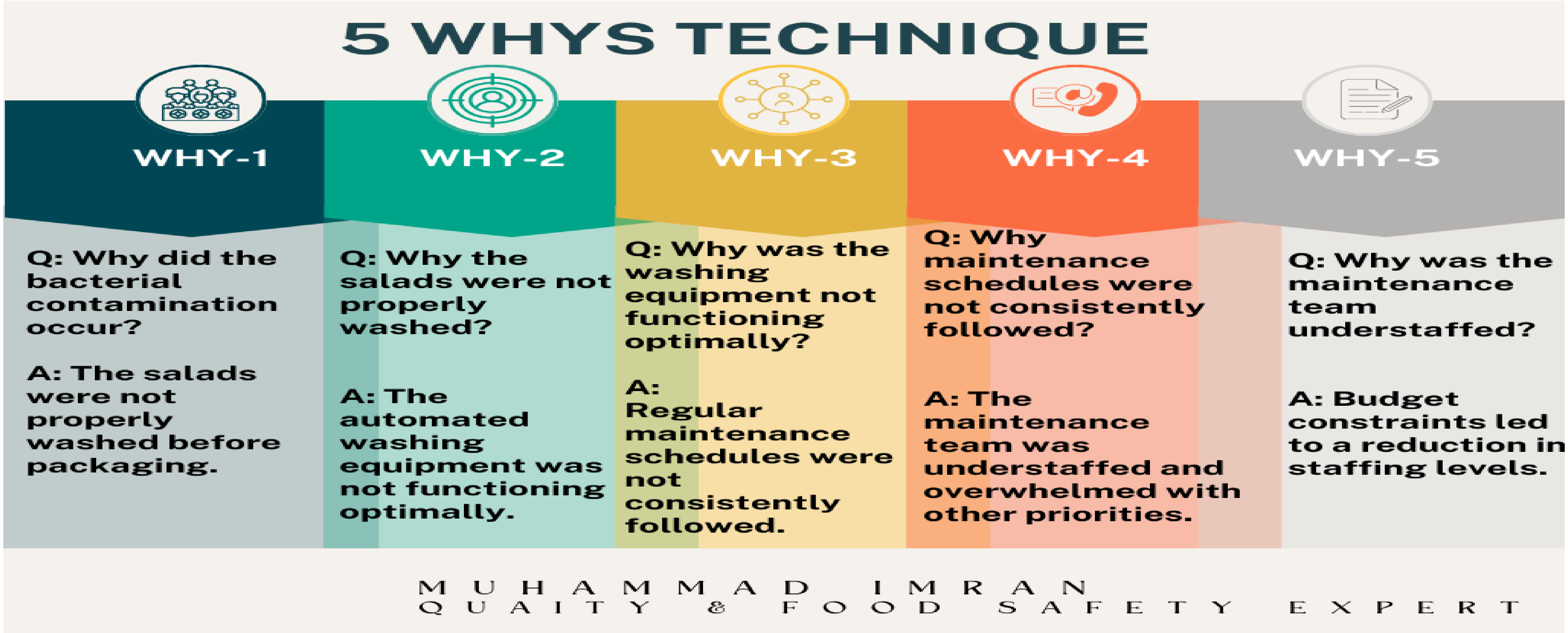
Root Cause Analysis (RCA)

- **Purpose:** Identify the fundamental cause of a problem, rather than just addressing its symptoms.
- **Methods:**
  - **5 Whys:** Keep asking "Why?" until the root cause is identified.
  - **Fishbone Diagram (Ishikawa):** Visualizes the cause-and-effect relationship of problems.

Toyota & 5 Whys Tool”

- This method was initially developed by Sakichi Toyoda; the founder of Toyota industries and till time is being used in Toyota Motor Corporation. This is considering as the basic tools for Toyota’s scientific approach.
- Along with Toyota this tool has widespread application in many other industries including food safety to rectify root cause of customer complaints and other non-conformances underlying in product and processing.

Problem: A Food Processing Facility Experiences An Increase In Bacterial Contamination In Its Packaged Salads, Leading To A Product Recall



Example for 5 Whys

Microsoft Word
Document

Activity: Identify the fundamental root cause of the problem using 5 Whys Tool:

Problem Statement: Choose ONE

1. Increasing drug abuse rates among college students in urban areas.
2. Rising cases of cyberbullying among teenagers on social media platforms.
3. Increasing cybercrimes and online money laundering
4. Lack of Financial Management among young adults
5. Visualize and explore trends in global climate change data over the past 50 years in Delhi NCR.
6. Visualize and explore trends in food and eating habits among college going youth.
7. Visualize and explore trends in Fitness level and Exercise patterns among youth.

SWOT

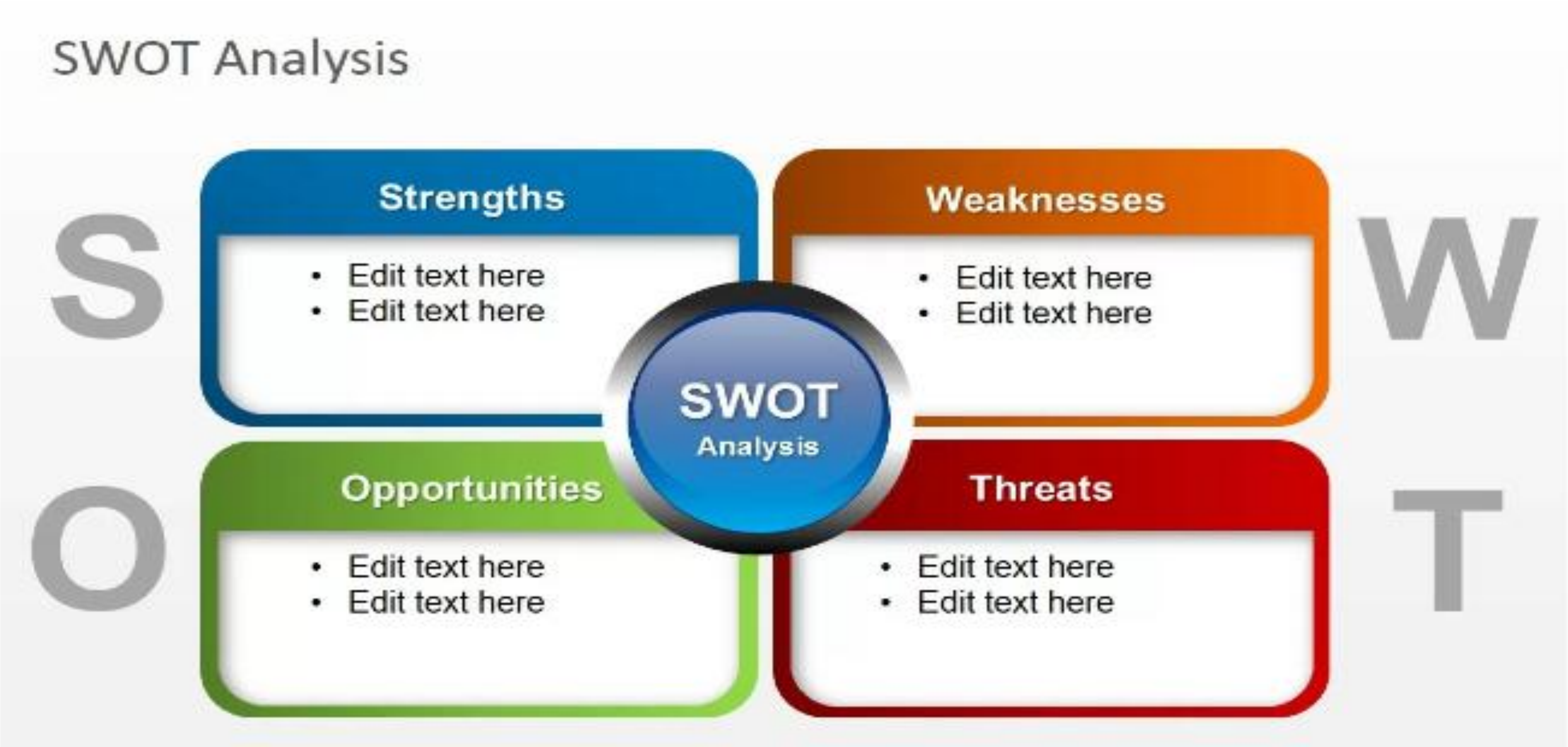
SWOT - Strengths and weaknesses, Opportunity and Threats

- SWOT (also known as SWOT matrix, SWOT analysis, and SWOT method) is a framework for identifying and analyzing a business's internal factors, namely strengths and weaknesses, and external factors, namely opportunities and threats. The framework helps differentiate and establish unique opportunities for companies within a broader market to determine their strategic business directions.

History of SWOT

- The history of SWOT analysis dates back to the 1960s. A management consultant from Stanford Research Institute, Albert Humphrey, and his team invented a framework to help businesses with more sustainable strategic planning.
- The framework was initially introduced as SOFT Analysis (Satisfactory, Opportunity, Fault, and Threat). After a decade of work, Albert and his team eventually proposed a SWOT analysis that assesses criteria such as products, processes, customers, distributions, finances, and administration.

SWOT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Understanding SWOT

- SWOT stands for Strengths, Weaknesses, Opportunities, and Threats. Strengths and weaknesses are internal factors within a company that can typically be controlled or changed. In contrast, opportunities and threats are external factors that companies cannot influence directly but can respond to strategically.
- **Strengths** describe what a business is good at, setting it apart from competitors.
- **Weaknesses** are barriers that hinder a company from performing at its best.
- **Opportunities** are external factors that could provide competitive advantages.
- **Threats** are potential elements that could harm the company.

In Short SWOT

STRENGTHS AND WEAKNESS ARE INTERNAL

OPPORTUNITY AND THREATS ARE EXTERNAL.

What is Personal SWOT Analysis?

- A personal SWOT analysis is a method of individual assessment. It can be done at any stage in life to determine self-improvement, educational choices, career paths, or career growth opportunities. You can use a personal SWOT for self-assessment or social comparison.

Why is Personal SWOT Analysis Important?

- When it comes to significant changes in your life, it involves gathering information, thinking, and analyzing. Conducting a personal SWOT analysis can help you avoid unforeseen mistakes because it requires you to address your strengths, weaknesses, opportunities, and threats.
- You can understand well the following aspects:
- **Strengths**: Recognizing your advantages over your competitors/peers and positioning yourself to achieve your goals.
- **Weaknesses**: Identifying weaknesses allows you to develop a plan to bridge your gaps.
- **Opportunities**: Identifying opportunities helps you determine chances that guide you toward your goals.
- **Threats**: Recognizing threats enables you to build a defensive plan against potential obstacles and unexpected challenges.

Quote

- *Being aware of your weaknesses could be your biggest strength – Gordon Hester.*

Assignment – 5 Marks

- SWOT Analysis on Individual (5 marks)
    1. Time management
    2. Health and Wellbeing
    3. Build Positive Attitude and behaviour
    4. Be Responsible
    5. Social Relationship
    6. Communication skills
    7. Mindful in reading Screentime

SWOT Analysis on Thematic Topic – 5 Marks

- Role of Youth being a Responsible Consumption in reducing greenhouse gas.

Root Cause Analysis

UNIT 2

What is RCA – Root Cause Analysis

It is a systematic process of discovering the origin of problems or the root cause of the problem in order to determine appropriate solutions. It is the fundamental aspect of problem solving in any field.

- Identify the underlying cause or causes of a problem rather than addressing the immediate symptoms. It ensure that same problem does not occur.

RCA – Root Cause Analysis

Step1: Problem must be clearly and understood [By gathering all data, documenting the data, understanding its impact on the process or system)

Step 2: Asking series of Why Questions – A method known as 5 Whys. The goal of asking this 5 Whys is to peel the symptoms and find out the root cause.

Step 3: One the root cause is been identified solutions can be addressed

Step 4: Develop solutions

Step 5: Implement and Monitor the Solutions

- Root Cause Analysis is not only just fixing problem but about
- Learning Opportunity
- Document Valuable Insights.
- How systems works and how it can be improved.
- Culture of Learning and Improvement

RCA – Root Cause Analysis

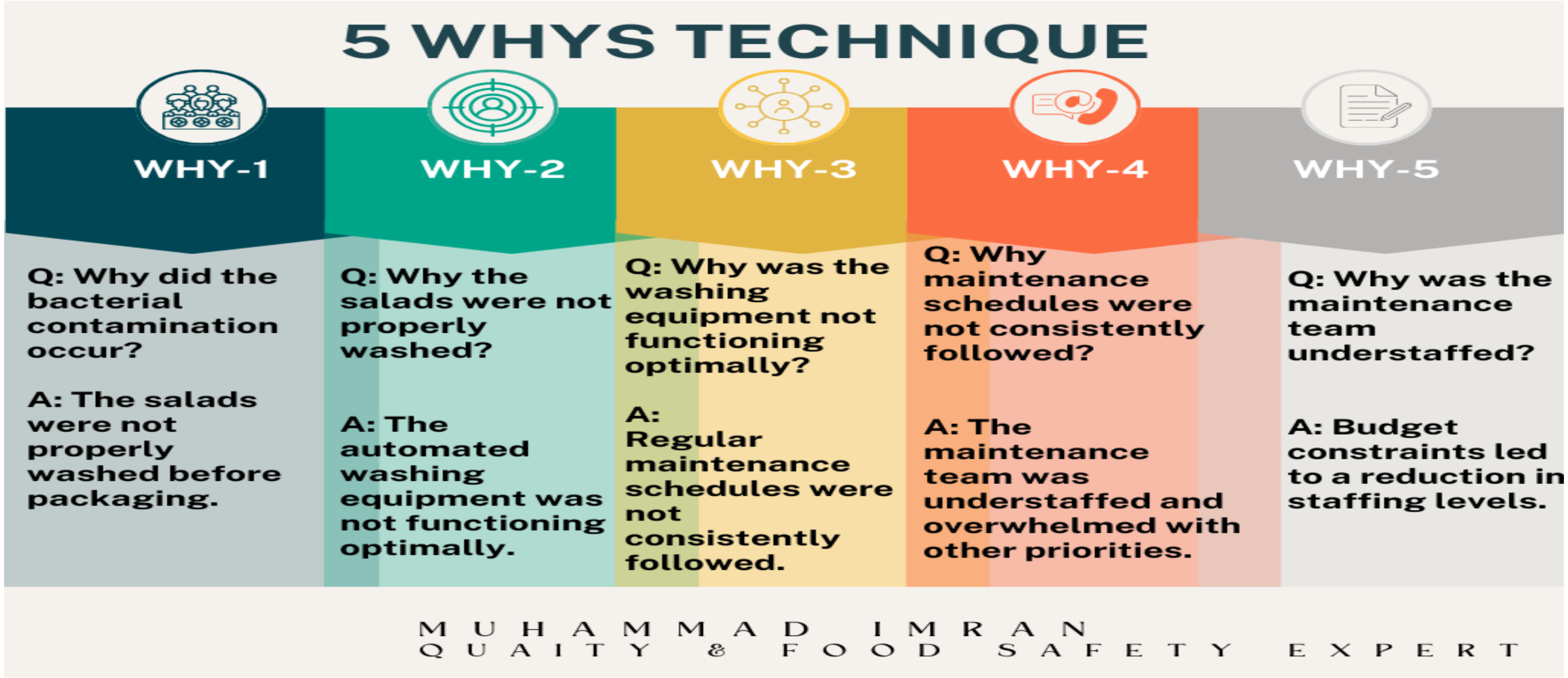
Root Cause Analysis (RCA)

- **Purpose:** Identify the fundamental cause of a problem, rather than just addressing its symptoms.
- **Methods:**
  - **5 Whys:** Keep asking "Why?" until the root cause is identified.
  - **Fishbone Diagram (Ishikawa):** Visualizes the cause-and-effect relationship of problems.

Toyota & 5 Whys Tool”

- This method was initially developed by Sakichi Toyoda; the founder of Toyota industries and till time is being used in Toyota Motor Corporation. This is considering as the basic tools for Toyota’s scientific approach.
- Along with Toyota this tool has widespread application in many other industries including food safety to rectify root cause of customer complaints and other non-conformances underlying in product and processing.

Problem: A Food Processing Facility Experiences An Increase In Bacterial Contamination In Its Packaged Salads, Leading To A Product Recall



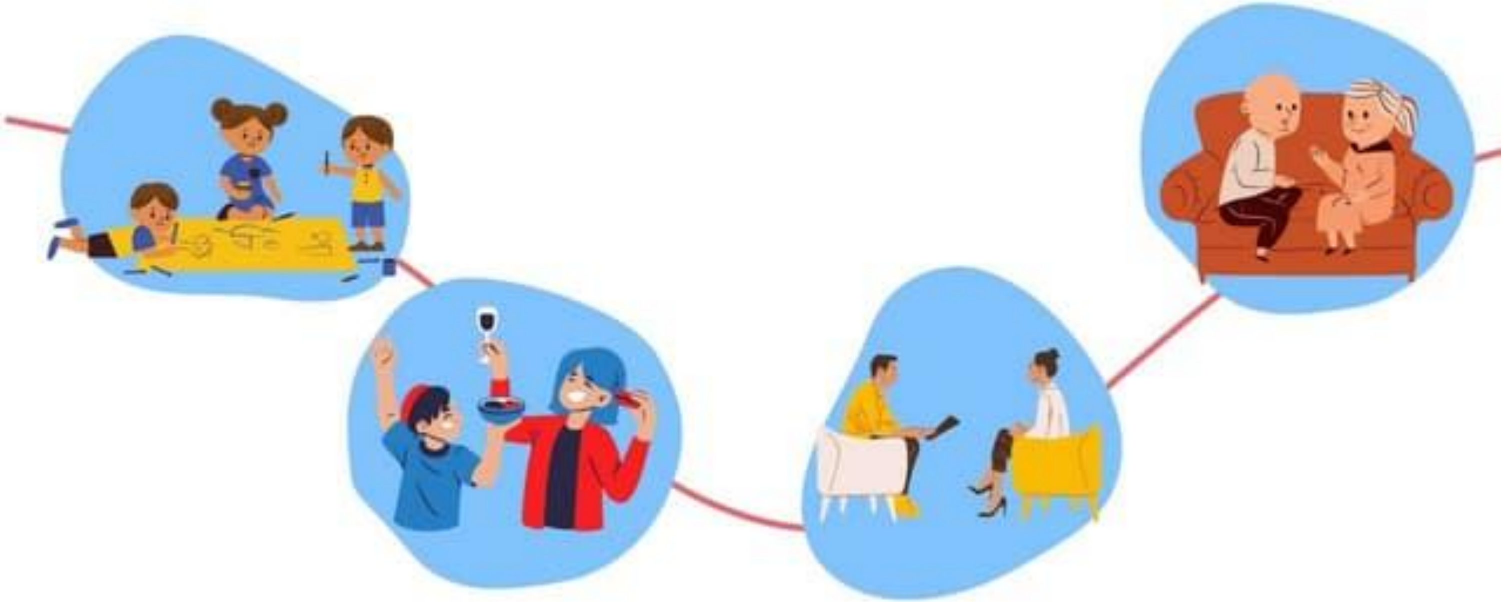
Example for 5 Whys

Microsoft Word
Document

Activity: Identify the fundamental root cause of the problem using 5 Whys Tool:

Problem Statement: Choose ONE

1. Increasing drug abuse rates among college students in urban areas.
2. Rising cases of cyberbullying among teenagers on social media platforms.
3. Increasing cybercrimes and online money laundering
4. Lack of Financial Management among young adults
5. Visualize and explore trends in global climate change data over the past 50 years in Tamil Nadu.
6. Visualize and explore trends in food and eating habits among college going youth.
7. Visualize and explore trends in Fitness level and Exercise patterns among youth.



SOCIALIZATION

What is Socialization?

Socialization is the lifelong process of learning norms, values, and behaviors necessary for societal interaction.

Socialization shapes identity, personality, and relationships.

Importance of socialization for students

- **Personal growth**: Develops self-concept, self-identity and confidence. Critical for personal development
- **Academic success**: Encourages collaboration and communication.
- **Career readiness**: Prepares students for professional settings.
- **Cultural adaptation**: Helps students integrate into diverse social settings.
- **Role in society**: Promote social order, social skills. understanding of societal roles and expectations

Agents of socialization

- **Family**: Primary agent during early life, shaping values, norms, and personality.
- **Education**: Teaches academic knowledge. Respect for authority. Shapes knowledge, discipline, and social skills.
- **Peer groups**: Influence friendships and identity. Promotes independence and identity exploration
- **Media**: Affects aspirations and awareness. Influences public opinions and social trends
- **Campus culture**: University traditions and student organizations.

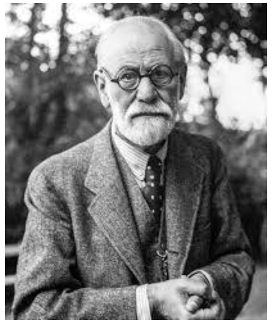
Types of socialization

- **Primary Socialization**: Occurs in early childhood. Early learning from family.
- **Secondary Socialization**: Continues through life. Adjusting to new roles in personal and academic life.
- **Anticipatory Socialization**: Preparing for future roles (e.g., internships, role models).
- **Resocialization**: Adapting to new environments.

Principles of socialization

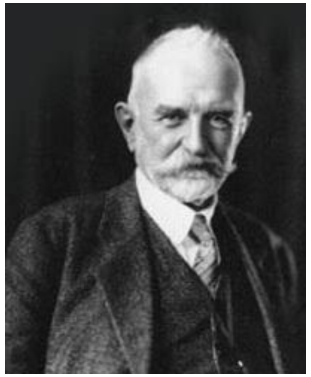
- **Internalization**: Absorbing university values.
- **Imitation**: Learning from role models.
- **Role learning**: Understanding social roles.
- **Feedback**: Adjusting behaviors based on peer and faculty input.

Theories of Socialization in College



Sigmund Freud: Psychoanalytic theory: socialization shapes the id, ego, and superego. Balancing personal desires with societal expectations.

George Herbert Mead: Symbolic interactionism: self-concept develops through social interaction and role-taking. Developing self-concept through peer interactions.



Jean Piaget: Cognitive development theory: children go through stages of learning social rules. Learning social norms through cognitive development.

Cultural variations in socialization

- **Individualism vs. Collectivism**:

Western cultures emphasize individual achievement, while Eastern cultures focus on collective well-being. Personal achievement vs. group harmony.

- **Gender roles**:

Expectations based on cultural norms. Gender-specific socialization shapes roles, expectations, and behaviors.

- **Diversity in campus life**:

Different experiences for international and regional students.

Impact of socialization on student behavior

- **Positive Effects**: Encourages adaptability and integration.
- **Negative Effects**: Can reinforce stereotypes and peer pressure.

Challenges in socialization for students

- **Cultural adjustment**: Language and tradition barriers.
- **Impact of technology**: Social media vs. real-life interactions.
- **Mental health concerns**: Pressure to conform or fit in.

Student Activity: Socialization Simulation Challenge

- Objective: To understand how social norms and behaviors are learned.
- Instructions:
- Form small groups and assign each group a social scenario (e.g., first day at college, internship, studying abroad).
- Discuss how socialization plays a role in the scenario.
- Present findings and reflect on challenges and adaptations.

- First day at college: How do students navigate new environments and friendships?
- Internship experience: How do students adapt to workplace culture?
- Studying abroad: How do students adjust to different cultural norms and expectations?
- Joining a student club: How do students learn to engage with diverse groups?

Conclusion

- Socialization in college influences academic success, social skills, and personal development. Understanding and adapting to new roles can create a more inclusive university culture.